# Irem **Topal**

## Physics Engineer | PhD Candidate

+(90) 539 463 8815 irem.topal@khas.edu.tr
Sisli , Istanbul, Turkey github.com/lffrrnt
lffrrnt.github.io

## Education

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|---|---|
| 2019 - present | PhD in Computational Science and Engineering, Kadir Has University, Istanbul, Turkey. |
| 2019 | PhD in Physics Engineering, Istanbul Technical University, Courses Transferred. |
| 2017 | MSc in Physics Engineering, Istanbul Technical University. |
| 2014 | BSc in Physics Engineering, Istanbul Technical University. |

## Research and Experience

**Present – October 2019** | **Research Assistant, Kadir Has University, Istanbul, Turkey.**
PhD funded by TUBITAK

**Sept 2021 – July 2021** | **Visiting PhD Student, International Center for Theoretical Physics, Trieste, Italy.**
Quantitative Life Sciences Section

**October 2019 – November 2018** | **Physics Tutor, Bahcesehir, Istanbul, Turkey.**
Worked one-on-one with undergraduate engineering students to prepare them for the exams in Physics I and II.

**October 2019 – April 2018** | **Data Scientist, Miletos Inc., Istanbul, Turkey.**

**Sept 2013 – August 2013** | **Engineering Intern, Eczacibasi-Monrol Nuclear Products, Kocaeli, Turkey.**

**August 2013 – July 2013** | **Intern, Istanbul University, Cerrahpasa Faculty of Medicine, Istanbul, Turkey.**
Received Nuclear Medicine Physics Certificate.

## Publications

2022 Topal, I., Eroglu, D., *"Reconstructing network dynamics of weakly coupled chaotic units from data"* (*under review in Physical Review Letters*)

2019 Topal, I., Servantie, J., *"Molecular Dynamics Study of the Thermal Conductivity in Nanofluids"*, Chemical Physics **516**, 147-151, (2018), DOI :`https://doi.org/10.1016/j.chemphys.2018.09.001`

## Talks and Posters

2021 *Reconstructing network dynamics of weakly coupled chaotic units from data* (Talk), Inverse Network Dynamics, Max Planck Institute for the Physics of Complex Systems, Dresden, Germany.

2021 *Network Reconstruction from Data* (Poster), The EPS conference on Statistical Physics of Complex Systems, ICTP, Trieste.

2021 *Reconstructing Network Dynamics of Weakly Coupled Chaotic Units* (Online), SIAM Conference on Applications of Dynamical Systems.

2020 *Sensing Network Dynamics of Weakly Coupled Chaotic Units* (Online), Dynamics Days Digital 2020.

2018 *Prediction of DNA Binding Sites for a Transcription Factor* (Poster), Machine Learning Summer School at Bosphorus (BYOYO), Istanbul.

2018 *Molecular Dynamics Study of the Thermal Conductivity in Nanofluids* (Talk), The annual Statistical Physics Days, İzmir.

## Skills

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| **Programming** | Mathematica, Maple, Julia |
| **Software** | VMD, Inkscape, LaTex, Grace |
| **Development tools** | git |
| **Operating systems** | Ubuntu, Mac OS X, Windows |

## Programming Languages

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|---|---|
| **Python** | ●●●●● |
| **Fortran** | ●●●●○ |
| **MATLAB** | ●●●●○ |

## Languages

- Turkish (Native)
- English

## Outreach and Volunteering

2013 İstanbul Technical University, Students' Representative Council, Undergraduate Student Representative of Physics Engineering